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REG.NO:

SCHOOL OF ADVANCED SCIENCES
DEPARTMENT OF MATHEMATICS
CONTINUOUS ASSESSMENT TEST - II
FALL SEMESTER 2024-2025

SLOT: C1+TC1+TCC1

Programme Name & Branch : B.Tech
Course Code and Course Name : BMAT205L - Discrete Mathematics and Graph Theory
Faculty Name(s) : Prof. M C Saravanarajan, Prof. D Ezhilmaran, Prof. G Deepa, Prof. A Manimaran, Prof. B Venkateswarulu, Prof. M Nalliah, Prof. Niranjan Hari, Prof. S Jagatheswari, Prof. Amrit Das, Prof. V Vijayakumar, Prof. V Murugan, Prof. Ankush Chanda, Prof. A Parthiban, Prof. Juhi Baroi, Prof.M Jagadeeswari, Prof. Sumit, Prof. Praveen
Class Number(s) : VL2024250102540, 2539, 2545, 2537, 2541, 2542, 2543, 2544, 2554, 2546, 2549, 2547, 2548, 2552, 2553, 2558, 2555
Date of Examination : 15/10/2024
Exam Duration : 90 minutes **Maximum Marks: 50**

ANSWER ALL THE QUESTIONS

Q. No	Question	M	CO	BL
1.	An encoding function $e: B^3 \to B^6$ is given by the generator matrix $\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$. a) Determine all code words generated by the matrix. b) Find the associated parity check matrix H c) Use H to decode the following received words 101100, 010100	10	2	2
2.	a) How many integers between 357 and 573 (both inclusive) are divisible by (i) at least one of 3, 5, 7? (ii) 7 and 5 but not 3? (iii) 7 but neither 3 nor 5?	5	3	3
	b) How many different ways are there to choose 5 donuts from the 7 varieties such that there must be at least 3 varieties? Assume that there are at least 3 donuts of each variety.	5		

3.	Solve the recurrence relation using generating functions $4a_{n-2} - 4a_{n-1} + a_n = 4^n,\ \ n \geq 2, a_0 = 2, a_1 = 8$	10	3	3
4.	Draw the Hasse diagram for the poset $\{S_{105}, D\}$, divisors of 105 under the relation divides. Hence or otherwise prove that it is a lattice, if so find two sub lattices and two non sub lattices. Also find $7 * (21 \oplus 5)$ and $5 \oplus (21 * 7)$	10	4	3
5.	Obtain the sum of products and product of sums canonical form of the Boolean expression $x_2' \oplus ((x_3' \oplus x_1 \oplus (x_2 * x_3)') * (x_3 \oplus (x_1' * x_2)))$	10	4	2

CAT II C1

1. a) $e(000) = (000000)$, $e(001) = (001011)$
$e(010) = (010111)$, $e(011) = (011100)$
$e(100) = (100110)$, $e(101) = (101101)$
$e(110) = (110001)$, $e(111) = (111010)$

b) $H = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$

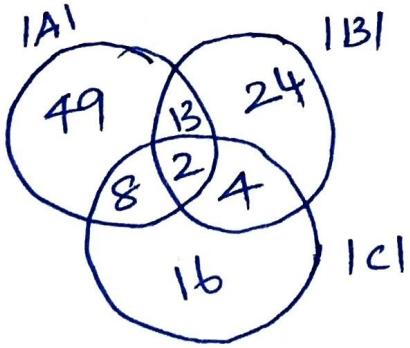
c) $r = 101100$, $Hr^t = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ 6th column of H
Decoded word is 101101

$r = 010100$, $Hr^t = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ Decoded word is ~~010110~~ 011100

2. a) $|A|$ = no of integers divisible by 3 $= \left\lfloor \frac{573}{3} \right\rfloor - \left\lfloor \frac{357}{3} \right\rfloor = 72$

$|B| = 43$ $|A \cap B| = 15$ $|B \cap C| = 6$

$|C| = 30$ $|A \cap C| = 10$ $|A \cap B \cap C| = 2$



i) 116 ii) 4 iii) 16

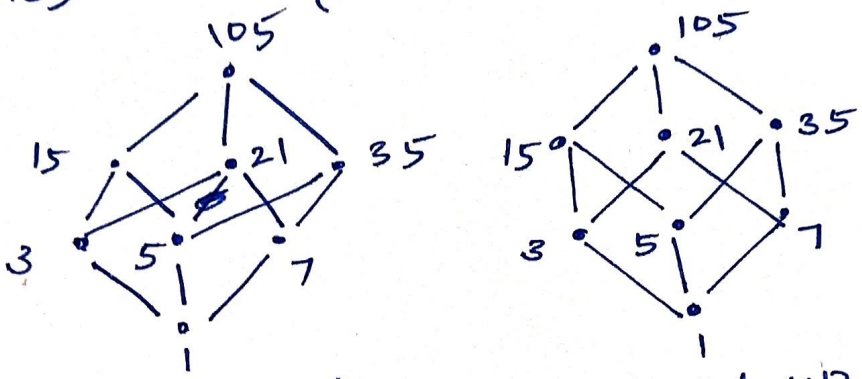
b) $n(7,3) \times n(7+2-1, 2) = 35 \times 21 = 735$

3. $G(x) = \dfrac{1+(1-4x)^2}{(1-4x)(1-2x)^2}$

$$= \frac{4}{1-4x} - \frac{2}{(1-2x)^2}$$

$$\therefore a_n = 4^{n+1} - (n+2)2^{n+1}$$

4. $\langle S_{105}, D \rangle = \{1, 3, 5, 7, 15, 21, 35, 105\}$



i) Every pair has GLB and LUB, ∴ lattice

ii) any two examples

iii) $7 \times (21+5) = 7 \quad 5+(21 \times 7) = 35$

5. SoP:

$x_1x_2x_3 + x_1x_2'x_3 + x_1'x_2'x_3 + x_1x_2'x_3' + x_1'x_2x_3' + x_1'x_2'x_3'$

PoS: $(x_1' + x_2' + x_3)(x_1 + x_2' + x_3')$